

1. Learning setup and guarantees

1. Statistical learning objects

A batch problem declares a domain $\mathcal{X}$, labels $\mathcal{Y}$, a hypothesis class $\mathcal{H}$ of maps $h : \mathcal{X} \rightarrow \mathcal{Y}$, a loss $\ell(h(x), y)$, a distribution $\mathcal{D}$ on (x, y) , and an ordered i.i.d. sample $S = ((x_i, y_i))_{i=1}^m \sim \mathcal{D}^m$.

For binary classification, $\mathcal{Y} = \{0, 1\}$ and zero-one loss is

$$\ell(h(x), y) = \mathbf{1}_{\{h(x) \neq y\}}.$$

$\mathcal{H}$ and ℓ are modeling choices; $\mathcal{D}$ is fixed but may be unknown. The sample is random before drawing and observed afterwards.

2. Population risk vs empirical risk

For $h \in \mathcal{H}$,

$$R(h) = E_{(X,Y) \sim \mathcal{D}}[\ell(h(X), Y)], \quad \hat{R}_{S(h)} = \frac{1}{m} \sum_{i=1}^m \ell(h(x_i), y_i).$$

$R(h)$ averages over a fresh population draw; $\hat{R}_{S(h)}$ averages over the observed sample. Under zero-one loss these are true misclassification probability and training-error fraction.

Pitfall: small $\hat{R}_{S(h)}$ is evidence about, not equality with, small $R(h)$. Keep both the hat and sample subscript.

3. ERM, realizability, and PAC

ERM returns any

$$\hat{h}_S \in \arg \min_{h \in \mathcal{H}} \hat{R}_{S(h)}.$$

Ties are allowed; the minimized quantity is sample risk.

Realizability means $\exists h^* \in \mathcal{H} : R(h^*) = 0$. Observed zero training error alone does not establish it.

$\mathcal{H}$ is realizably PAC learnable if one learner A and $m_{\mathcal{H}} : (0, 1)^2 \rightarrow \mathbb{N}$ exist such that, for every $(\varepsilon, \delta) \in (0, 1)^2$, every realizable $\mathcal{D}$, and every $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$,

$$P_{S \sim \mathcal{D}^m}(R(A(S)) \leq \varepsilon) \geq 1 - \delta.$$

ε is risk tolerance; δ is failure probability. Probability is over S and any learner randomness.

2. Finite-class control

1. Fixed hypothesis: Hoeffding

If h is fixed independently of the i.i.d. sample and $\ell \in [0, 1]$, then for $\gamma > 0$,

$$P(|\hat{R}_{S(h)} - R(h)| > \gamma) \leq 2 \exp(-2m\gamma^2).$$

The bounded loss supplies the range; sample independence supplies the product structure.

Selection warning: $\hat{h}_S$ depends on S . Substituting it for fixed h in this statement changes an assumption. Protect data-dependent selection by controlling all candidates simultaneously.

2. Simultaneous deviation over finite $\mathcal{H}$

For fixed h , let

$$E_{h(\gamma)} = \{|\hat{R}_{S(h)} - R(h)| > \gamma\}.$$

If $\mathcal{H}$ is finite, S is i.i.d., and $\ell \in [0, 1]$, then

$$P\left(\bigcup_{h \in \mathcal{H}} E_{h(\gamma)}\right) \leq 2|\mathcal{H}| \exp(-2m\gamma^2).$$

Thus, with probability at least $1 - \delta$,

$$\forall h \in \mathcal{H}, \quad |\hat{R}_{S(h)} - R(h)| \leq \sqrt{\frac{\ln\left(2\frac{|\mathcal{H}|}{\delta}\right)}{2m}}.$$

Equivalently, deviation at most γ is ensured by

$$m \geq \frac{\ln\left(2\frac{|\mathcal{H}|}{\delta}\right)}{2\gamma^2}.$$

No independence between the events E_h is required.

3. What the simultaneous event gives ERM

Let $\hat{h}_S$ minimize empirical risk and let $h^* \in \arg \min_{h \in \mathcal{H}} R(h)$. If every deviation is at most γ ,

$$\begin{aligned} R(\hat{h}_S) &\leq \hat{R}_{S(\hat{h}_S)} + \gamma \\ &\leq \hat{R}_{S(h^*)} + \gamma \\ &\leq R(h^*) + 2\gamma. \end{aligned}$$

The middle inequality is ERM. The first and last inequalities use the same simultaneous event, which remains valid for the sample-selected hypothesis.

Pitfall: deviation tolerance γ yields an excess-risk term 2γ .

4. Sharper realizable, consistent route

With zero-one loss, call h **consistent** if $\hat{R}_{S(h)} = 0$. For fixed h with $R(h) > \varepsilon$,

$$P(h \text{ is consistent}) = (1 - R(h))^m \leq \exp(-m\varepsilon).$$

If $\mathcal{H}$ is finite, realizability holds, and ERM returns a consistent hypothesis, then

$$P(\exists h : R(h) > \varepsilon \text{ and } \hat{R}_{S(h)} = 0) \leq |\mathcal{H}| \exp(-m\varepsilon).$$

It is sufficient that

$$m \geq \frac{1}{\varepsilon} \left(\ln|\mathcal{H}| + \ln\left(\frac{1}{\delta}\right) \right)$$

to guarantee $P(R(\hat{h}_S) \leq \varepsilon) \geq 1 - \delta$.

This $\frac{1}{\varepsilon}$ rate uses realizability and consistency. It is not the two-sided $\frac{1}{\gamma^2}$ deviation result. Neither sample bound proves an ERM implementation is computationally efficient.

5. Which finite-class bound?

- Fixed h + i.i.d. bounded loss $\rightarrow$ Hoeffding.
- After selection: finite bounded-loss $\mathcal{H} \rightarrow$ simultaneous control and 2γ excess; add zero-one realizability and consistency $\rightarrow$ direct $\frac{1}{\varepsilon}$ risk rate.

Sample complexity $\neq$ computational running time.

3. Finite-sample behaviors

1. Restriction and growth function

For distinct $C = \{x_1, \dots, x_m\}$,

$$\mathcal{H}|_C = \{(h(x_1), \dots, h(x_m)) : h \in \mathcal{H}\} \subseteq \{0, 1\}^m.$$

Count distinct label vectors, not hypotheses or parameter values. The growth function is

$$\Pi_{\mathcal{H}}(m) = \max_{C \subset \mathcal{X}: |C|=m} |\mathcal{H}|_C|.$$

For a nonempty binary class, $1 \leq \Pi_{\mathcal{H}}(m) \leq 2^m$. A count on one set is only a lower bound on the maximum unless every m -point set is also bounded.

For thresholds on three ordered points, infinitely many threshold values collapse to the four restrictions 000, 001, 011, 111. The restriction count, not the parameter count, is the finite-sample object.

2. Shattering and VC dimension

$\mathcal{H}$ **shatters** C when

$$\mathcal{H}|_C = \{0, 1\}^{|C|},$$

so every binary labeling of that fixed set is realized. Missing even one vector prevents shattering.

$$\text{VCdim}(\mathcal{H}) = \max\{|C| : \mathcal{H} \text{ shatters } C\},$$

or ∞ if arbitrarily large finite sets are shattered.

To prove $\text{VCdim}(\mathcal{H}) = d$:

- **lower bound:** exhibit one d -point set and realize all 2^d labelings;
 - **upper bound:** prove every $(d+1)$ -point set has a missing labeling.
- Failure on one $(d+1)$ -point set does not prove the upper bound.

3. Interval classifiers: VC dimension two

Let $\mathcal{I} = \{h_{a,b}(x) = 1_{\{a \leq x \leq b\}} : a \leq b\}$. Two ordered points $x_1 < x_2$ are shattered: place an interval away from both, around either point, or across both to realize 00, 10, 01, 11.

For every ordered triple $z_1 < z_2 < z_3$, labeling 101 is impossible: an interval containing z_1 and z_3 must contain z_2 . Therefore

$$\text{VCdim}(\mathcal{I}) = 2.$$

This is one existential two-point construction plus a universal three-point obstruction.

4. Sauer-Shelah and learning

1. Sauer-Shelah lemma

If $\text{VCdim}(\mathcal{H}) \leq d < \infty$, with d a nonnegative integer, then for every nonnegative integer m ,

$$\Pi_{\mathcal{H}}(m) \leq \sum_{i=0}^d \binom{m}{i},$$

where $\binom{m}{i} = 0$ for $i > m$. For fixed finite d , this is polynomial in m , while the unrestricted binary count is 2^m .

The lemma controls finite-sample behavior; $\mathcal{H}$ itself may be infinite. For intervals on the five ordered points in the Tutorial, there is one all-zero vector and one for every nonempty consecutive positive block:

$$1 + 5 + 4 + 3 + 2 + 1 = 16, \quad \text{while} \quad 2^5 = 32.$$

The exact binomial sum gives a finite- m guarantee; $O(m^d)$ records only its fixed- d growth order.

2. Sauer-Shelah proof skeleton

Define $F(m, d)$ as the maximum restriction count over all binary classes with VC dimension at most d and all m -point sets. For a realized family V , delete the last coordinate: U contains all prefixes and $D \subseteq U$ those with both extensions. Then $|V| = |U| + |D|$.

Now $\text{VCdim}(U) \leq d$. If D shattered d old points, its two extensions would shatter $d+1$ points; thus $\text{VCdim}(D) \leq d-1$ and

$$F(m, d) \leq F(m-1, d) + F(m-1, d-1).$$

With $F(0, d) = F(m, 0) = 1$, induction closes using Pascal's identity

$$\binom{m-1}{i} + \binom{m-1}{i-1} = \binom{m}{i}.$$

3. Finite VC dimension: current conclusion

For binary zero-one classification, $d = \text{VCdim}(\mathcal{H}) < \infty$ implies distribution-free realizable PAC learnability. A standard ERM upper bound has order

$$m_{\mathcal{H}}(\varepsilon, \delta) = O\left(\frac{d \ln(\frac{1}{\varepsilon}) + \ln(\frac{1}{\delta})}{\varepsilon}\right).$$

Big- O hides a constant, so this is not an exact certificate or runtime claim. Because restrictions depend on the random sample, do not blindly substitute $\Pi_{\mathcal{H}}(m)$ for $|\mathcal{H}|$ in the finite-class union bound. Agnostic equivalence and symmetrization remain Chapter 3 material.